

BASIC SIGHT-SINGING AND EAR TRAINING FOR CHURCH SINGERS

UNIT 3.2

PITCH

BASIC DEFINITIONS: THE CONCEPT OF “TONIC”

INTERVALS AND PATTERNS IN THE SOL-DO TETRACHORD



THE CONCEPT OF “TONIC”

Most relatively simple melodies, including those we encounter in Orthodox church music have what is called a “**tonal center**,” known in musical terminology as a **tonic**. The pitch that serves as the tonic of a given melody has a type of “gravitational attraction” that affects the notes around it, as if “pulling” them towards itself.

Simple melodies often begin on the tonic, and they almost always, with very few exceptions, end on the tonic. The ear experiences a sense of “arrival” or “repose” when melody comes to rest on the tonic. (This phenomenon is even stronger in harmonic music, which will be considered in a later lesson.)

In Movable DO Solfege, the tonic, or the **point of reference**, in the melodies we have encountered thus far, has always been called “DO”: DO **is** the tonic, by definition. When DO is the tonic of a melody, it is said to be built on a **major scale** or to be in a **major key**.

THE STRUCTURE OF THE MAJOR SCALE OR

HOW CAN WE TELL IF A SCALE IS MAJOR?

The DO-FA Tetrachord is an important component of the major scale, constituting the first half of it. As defined in the Unit 2.1, it consists, by definition of the following pattern of intervals:

DO - RE A **Whole step** or a **Major Second**

RE - MI A **Whole step** or a **Major Second**

MI - FA A **Half step** or a **Minor Second**

This arrangement of pitches—**whole step, whole step, half step**—is what gives the major scale its distinctive character, or what we might call its “major-ness.” You can hear the character of the scale by the time you reach MI, the third note of the scale. The interval DO-MI, as we learned in Unit 2.1 (Slide 3), is a Major Third, and so a scale whose third note is MI—a Major Third above DO, the tonic, is very likely to be a **major** scale.

COMPLETING THE MAJOR SCALE: THE SOL-DO TETRACHORD

The DO-FA Tetrachord begins (and defines) the major scale. But the scale has 4 more notes, making a total of 8. The remaining component of the major scale is the SOL-DO Tetrachord, which, as it happens, consists of exactly the same pattern of intervals as DO-FA:

SOL - LA A **Whole step** or a **Major Second**

LA - TI^{*)} A **Whole step** or a **Major Second**

TI - DO A **Half step** or a **Minor Second**

The SOL-DO tetrachord differs, however, from the DO-FA tetrachord in one important respect: whereas the tonic, or reference note, of DO-FA is the **first** note of the tetrachord, in the SOL-DO tetrachord, the tonic (DO) is the **fourth** note of the tetrachord.

^{*)}In Russia and some Western European countries this note is called “SI”; but in English-speaking countries the syllable “TI” is used, so as not to confuse it with the letter-note-name “C.”

FEEL THE TONIC FUNCTION OF DO

It is important that you learn to experience the **tonic function** of DO, with regard to these two tetrachords. This is best achieved by singing DO-RE-MI-FA upwards from the tonic note DO, and by singing DO-TI-LA-SOL downwards from DO. Remember that in Moveable-DO Solfège, **any** pitch can serve as the tonic DO. The relationship of the pitches in the tetrachords will always remain in the same.



IMPORTANT EXERCISE!

1. Go to your piano or bring up the virtual piano keyboard here: [Virtual Piano - Online Piano Keyboard](#)
2. Play any note and call it DO.
3. Sing DO-RE-MI-FA upwards from DO and FA-MI-RE-DO, returning to the tonic.
4. Next, sing DO-TI-LA-SOL downwards from DO and SOL-LA-TI-DO, back to the tonic.

← Tonic →
SOL - LA - TI - **DO** - RE - MI - FA

BUILDING THE COMPLETE MAJOR SCALE

Arranging the two tetrachords one after the other yields the complete major scale, which begins on DO and ends on DO.

DO - RE - MI - FA - SOL - LA - TI - **DO**

The interval between FA and SOL is a whole step, which causes the major scale, **by definition**, to have the following sequence of intervals: (W=whole step, H=half step)

W W H W W W H
DO - RE - MI - FA - SOL - LA - TI - **DO**

The pitch relationships in the major scale are not accidental or coincidental. They are based on mathematical ratios and laws of physics and acoustics. These were first “discovered” by the Ancient Greek mathematician Pythagoras (6th century B. C.)

INTERVALS IN THE SOL-DO TETRACHORD

In Slide 4 above, we saw that in terms of intervals the structure of the SOL-DO Tetrachord is the same as that of the DO-FA Tetrachord. Adding the four notes SOL, LA, TI, DO to our array of possible pitches creates some additional intervals that we need to identify and learn. The downward arrow signifies that second note is below the first one.

DO - TI↓	Half Step (also known as a Minor Second)
DO-LA↓	Minor Third (formed by stepping down a half step and a whole step)
DO - SOL↓	Perfect Fourth (formed by stepping down a half step and 2 whole steps)
TI-SOL↓	Major Third (formed by stepping down 2 whole steps)

MORE INTERVALS

Additionally, we can form intervals between the notes of the two tetrachords. For example,

MI-SOL & TI-RE

Minor Thirds (formed by stepping a whole step and a half step)

FA - LA

Major Third (formed by stepping 2 whole steps)

RE-SOL, MI-LA
LA-RE & TI-MI

Perfect Fourths (formed by stepping 4 notes—a total of 2 whole steps and a half step)

RE-LA, MI-TI, FA-DO,
SOL-RE & LA-MI

Perfect Fifths (formed by stepping a total of 5 notes—a total 3 whole steps and a half step)

We will explore the full array of intervals that can be formed from the notes of the complete major scale in the next unit. The intervals presented above will be featured in the Pitch Exercises 3.1, 3.2, & 3.3.

MORE PITCH PATTERNS IN ORTHODOX CHANT

Now that you have been introduced to the entire Major Scale, which includes both the DO-FA and the SOL-DO tetrachords, you can sing a number of **additional** melodies in the Eight Tones. In some melodies, the note SOL above the FA is used. In others, the TI and LA below DO are used.

Using the “BEST PRACTICES” we introduced earlier (Unit 1.1, Slide 10)

1. Set the pulse and read through the rhythm FIRST!
2. Analyze the Sol-Fa syllables in each line. Note similarities and differences.
3. Keeping the pulse, read the Sol-Fa syllables, trying to hear the pitch in your mind.
4. Sing on the Sol-Fa syllables, keeping a steady pulse.
5. The pattern of phrases is indicated by numbers enclosed in circles.
6. Whenever you're ready (before or after practicing Pitch Exercises 3.1, 3.2 and 3.3), record yourself singing these examples.

RESURRECTIONAL TROPARION

TONE 2 (COMMON CHANT)

① RE RE RE MI MI ——— MI FA SOL FA MI
When You de-scend - ed to death, O Life Im - mor - tal,

② MI ——— MI RE DO
You slew hell with the splendor of Your God - head.

① RE MI MI MI MI FA SOL FA MI
And when from the depths You raised the dead,

② MI ——— MI RE RE RE DO
all the powers of heav - en cried out:

Slur connecting two notes.

Here we see two alternating phrases, the first using the notes RE, MI, FA, SOL, and the second—the notes MI, RE, DO. In each phrase these notes are adjusted to accommodate the syllables of text, but the **pattern** of pitches remains the same. When two or more notes are sung to the same syllable of text, they are connected by a **slur**. This pattern is used for many other troparia and kontakia in Tone 2 of the so-called “common” Russian chant.

CATEGORIES OF MELODIC PATTERNS WITHIN THE EIGHT TONES

The system of Eight Tones used in the Eastern Slavic (Russian, Ukrainian, and their “daughter” Churches in English-speaking countries) comprises different melodic patterns for various categories of Orthodox hymns. Different patterns are used for:

1. **Troparia** and **Kontakia**, sung at the end of Vespers (aka Dismissal Troparion), the beginning of Matins, and after the Little Entrance at the Divine Liturgy.
2. **Stichera**, sung at “Lord, I call” and the Aposticha of Vespers, and the Praises of Matins.
3. **Heirmoi**, sung at the beginning of each ode of the Kanon at Matins.
4. **Prokeimena** and **Alleluia**, sung before the reading of Scripture at Vespers, Matins, and Divine Liturgy.

Within this general scheme there are variants: “Common” Chant, Kievan Chant, Znamenny Chant, Valaam Chant, and other regional varieties.

STICHERON (THEOTOKION DOGMATIKON)

TONE 3 (COMMON CHANT)

The musical notation is presented on three staves, each with a single line. Notes are represented by vertical stems with dots for heads. Circled numbers 1 and 2 indicate the start of alternating phrases. The first phrase consists of the notes FA, MI, and a long horizontal line. The second phrase consists of FA, MI, RE, DO, and a long horizontal line. The text is written below the notes, with hyphens indicating syllables that span multiple notes.

① FA FA FA FA FA MI ② FA _____ FA

How can we not won - der at Your mystical childbearing,

FA FA MI RE DO ① FA _____ FA FA FA MI

O Exalted Moth - er? For, without receiving the touch of man,

② FA _____ FA MI RE DO

you gave birth to a Son in the flesh, O Immaculate Vir - gin.

Like the Tone 2 Troparion pattern, Sticheron Tone 3 has two alternating phrases that are repeated as many times as the particular text requires. The first phrase consists of the notes FA-MI, while the second—FA-MI-RE-DO. The **Theotokion Dogmatikon** is a sticheron sung at the end of “Lord, I call,” which sets forth and praises the dogmas pertaining to the incarnation of Christ from the Most Holy Virgin Theotokos.

STICHERON (THEOTOKION DOGMATIKON)

TONE 7 (COMMON CHANT)

① FA FA FA MI MI FA SOL SOL SOL FA FA MI
No tongue can speak of your won-der-ful child - bear - ing

② MI MI SOL MI ————— MI RE RE DO
for the or - der of nature was over - ruled by God.

① FA FA FA MI FA SOL ————— SOL FA MI
You were re - vealed to be a mother above na - ture,

② MI MI MI SOL MI ————— MI RE DO
for you re-mained a virgin beyond reason and under-stand - ing.

Another sticheron Tone pattern that, like Tone 3, has two alternating phrases is Tone 7. Notice that in Phrase 2 there is a **leap** of a Minor Third (MI-SOL). Use your sense of **pitch memory** to remember these two notes from Phrase 1, which contains the movement MI-FA-SOL. Thinking of the FA between MI and SOL helps find the second note.

STICHERON (THEOTOKION DOGMATIKON)

TONE 5 (COMMON CHANT)

① RE RE RE MI DO DO ② MI ————— MI

In the Red Sea of old, a type of the Virgin Bride was pre -

MI RE ③ RE RE ——— RE TI DO TI LA ① RE RE ——— RE

fig-ured. There Moses divided the wa - ters; here Gabriel assisted

RE RE MI DO DO ② MI ————— MI MI MI

in the mir - a - cle. There Israel crossed the sea without get - ting

RE ③ RE RE ——— RE TI DO DO TI LA

wet. here the Virgin gave birth to Christ with - out seed.

Slightly more complex in structure is Sticheron Tone 5, which has three alternating phrases, built, respectively, on the pitches 1) RE-MI-DO, 2) MI-RE, and 3) RE-TI-DO-TI-LA. Intervals of the third—both major and minor—are used prominently here: Major—MI-DO, and Minor—RE-TI and DO-LA (with TI in between).

RESURRECTIONAL TROPARION

TONE 5 (COMMON CHANT)

① RE RE RE RE MI DO DO DO ② MI—
 Let us, the faithful praise and wor - ship the Word, co-eternal

MI MI RE ③ RE RE RE TI DO TI LA
 with the Father and the Spir-it, born for our salvation from the Vir - gin

① RE RE RE RE RE MI DO DO DO ② MI MI MI RE
 for He willed to be lifted up on the Cross in the flesh, to en-dure death,

③ RE TI DO TI LA Cad. MI MI RE DO DO DO TI DO TI LA
 and to raise the dead by His glo - rious Res - ur - rec - tion.

For every rule there is an exception. What do you notice about Troparion Tone 5 in relation to Sticheron Tone 5 on the preceding slide?

STICHERON (THEOTOKION DOGMATIKON)

TONE 8 (COMMON CHANT)

The musical notation for Sticheron Tone 8 is presented on a single staff with a C-clef. The notes are represented by dots with stems, and the intervals are indicated by horizontal lines. The text is written below the staff, with hyphens indicating where a phrase is repeated.

Phrase 1: MI FA MI FA MI RE (circled 1)

Phrase 2: FA ——— FA FA SOL FA MI (circled 2)

Phrase 3: FA FA FA MI RE FA MI RE DO (circled 3)

Phrase 1a: MI ——— MI (circled 1a)

Text Application:

The King of Heav - en, because of His love___ for man,
 ap-peared on earth___ and dwelt with men. He took flesh from the pure
 Vir - gin, and after assuming it, He came forth___ from her.
 The Son is one: in two na - tures, yet one Per - son.

Sticheron Tone 8, like Sticheron Tone 5, has three alternating phrases, built, respectively, on the pitches 1) MI-FA-MI-RE, 2) FA-SOL-FA-MI, and 3) FA-MI-FA-MI-RE-DO. Phrase 1a is a slight variation of Phrase 1, not repeating the opening MI-FA. Noticing all these details will help you sing these Tone patterns to a huge variety of texts that come up during the liturgical year.

You are now ready to explore and master the
intervals in the
SOL-DO TETRACHORD

Please proceed to
PITCH EXERCISES 3.1, 3.2, & 3.3

The instructions for these exercises are
contained in the video files and should be
self-explanatory.

